MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain distractor. This chapter format does not reproduce the official format of the EAM.

1. $e^0 =$
   - A. 0
   - B. 1
   - C. e
   - D. -1
2. $e^x e^y =$
   - A. e^{x+y}
   - B. e^{xy}
   - C. $e^x + e^y$
   - D. e^{x-y}
3. Derivative of e^{-2x} :
   - A. e^{-2x}
   - B. $2e^{-2x}$
   - C. $-e^{-2x}$
   - D. $-2e^{-2x}$
4. e^x = 1 is equivalent to
   - A. $x = 1$
   - B. $x = e$
   - C. $x = 0$
   - D. no solution
5. The sign of e^{-x} is
   - A. negative if x>0
   - B. Always strictly positive
   - C. zero at 0
   - D. Variable

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain distractor. This chapter format does not reproduce the official format of the EAM.

1. $e^{2x} < e^3$ is equivalent to
   - A. $x < 3$
   - B. $x > \frac{3}{2}$
   - C. $x < \frac{3}{2}$
   - D. $x < \sqrt{3}$
2. $u_n = e^{3n}$ has common difference/ratio
   - A. e^3
   - B. 3

C. e

D. $3e$

3. Tangent at e^x at 0:

A. $y = x$

B. $y = ex$

C. $y = 1$

D. $y = x + 1$

4. Derivative of xe^x :

A. e^x

B. $(x + 1)e^x$

C. xe^x

D. xe^{x-1}

5. $e^x \geq 1 + x$ Equality

A. for every x

B. at 1 only

C. at 0 only

D. Never

True or false? Justify.

1. $e^{2x} = 2e^x$ for all x.

2. The function $t \mapsto e^{-3t}$ is decreasing.

3. A continuous rate k=0.1 corresponds exactly to 10% increase in one unit of time.

Personal review

I keep the power rules; I know positivity; I differentiate with the factor; I separate
model, exact calculation and simulation.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.